

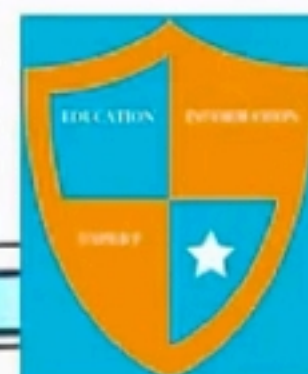
A woman is standing outdoors, wearing a dark blue long-sleeved tunic with a patterned vest over it. She is holding a red smartphone in her right hand. The background shows a brick wall and some greenery.

Rukhsar Kanwal

Mth631

Best Of luck

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \dots + m_n x_n$ for some constants m_i 's is called ———



page 112

Answer (Please select your correct option)

☐ linear function

☐ quadratic function

☐ constant function

☐ None of these



EDUCATION INFORMATION EXPERT



2.17.1 Linear Function

A *linear function* is a function of the form

$$L(\mathbf{X}) = m_1x_1 + m_2x_2 + \cdots + m_nx_n,$$

where $m_1, m_2, \dots, m_n$ are constants. From definition of differentiable at $\mathbf{X}_0$ if and only if there is a linear function L such that f can be approximated so well near $\mathbf{X}_0$ by

$$L(\mathbf{X}) - L(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0)$$

MTH631 Real Analysis II

Question No : 9 of 25

For the function $f(x,y) = 3x^2y^3 + xy$ in two variables, $f_y(0,0) = \dots\dots\dots$



Answer (Please select your correct option)

- ☒ 1
- ☐ -1
- ☐ 0
- ☐ None of these

correct



it is an r th-order partial derivative of f . The function

$$f(x, y) = 3x^2y^3 + xy$$

has partial derivatives everywhere. Its first-order partial derivatives are

$$f_x(x, y) = 6xy^3 + y, \quad f_y(x, y) = 9x^2y^2 + x.$$

Its second-order partial derivatives are

$$\begin{aligned} f_{xx}(x, y) &= 6y^3, & f_{yy}(x, y) &= 18x^2y, \\ f_{xy}(x, y) &= 18xy^2 + 1, & f_{yx}(x, y) &= 18xy^2 + 1. \end{aligned}$$

There are eight third-order partial derivatives. Some examples are

$$f_{xxy}(x, y) = 18y^2, \quad f_{xyx}(x, y) = 18y^2, \quad f_{yxx}(x, y) = 18y^2.$$

Compute $f_{xx}(0, 0)$, $f_{yy}(0, 0)$, $f_{xy}(0, 0)$, and $f_{yx}(0, 0)$ if

$$f(x, y) = \begin{cases} \frac{(x^2y + xy^2) \sin(x-y)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

If $(x, y) \neq (0, 0)$, the ordinary rules for differentiation, applied separately to x and y , yield

$$\begin{aligned} f_x(x, y) &= \frac{(2xy + y^2) \sin(x-y) + (x^2y + xy^2) \cos(x-y)}{x^2 + y^2} \\ &\quad - \frac{2x(x^2y + xy^2) \sin(x-y)}{(x^2 + y^2)^2}, \quad (x, y) \neq (0, 0), \end{aligned}$$

and

$$\begin{aligned} f_y(x, y) &= \frac{(x^2 + 2xy) \sin(x-y) - (x^2y + xy^2) \cos(x-y)}{x^2 + y^2} \\ &\quad - \frac{2y(x^2y + xy^2) \sin(x-y)}{(x^2 + y^2)^2}, \quad (x, y) \neq (0, 0). \end{aligned}$$

These formulas do not apply if $(x, y) = (0, 0)$, so we find $f_x(0, 0)$ and $f_y(0, 0)$ from their definitions as difference quotients:

$$\begin{aligned} f_x(0, 0) &= \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0, \\ f_y(0, 0) &= \lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{0 - 0}{y} = 0. \end{aligned}$$

Setting $y = 0$ in (2.32) and (2.33) yields

$$f_x(x, 0) = 0, \quad f_y(x, 0) = \sin x, \quad x \neq 0,$$



$$f_y(0,0) = \lim_{y \rightarrow 0} \frac{f(0,y) - f(0,0)}{y} = \lim_{y \rightarrow 0} \frac{0 - 0}{y} = 0.$$

Setting $y = 0$ in (2.32) and (2.33) yields

$$f_x(x,0) = 0, \quad f_y(x,0) = \sin x, \quad x \neq 0,$$

2.16. Directional Derivative

so

$$f_{xx}(0,0) = \lim_{x \rightarrow 0} \frac{f_x(x,0) - f_x(0,0)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0.$$

$$f_{yx}(0,0) = \lim_{x \rightarrow 0} \frac{f_y(x,0) - f_y(0,0)}{x} = \lim_{x \rightarrow 0} \frac{\sin x - 0}{x} = 1.$$

Setting $x = 0$ in (2.32) and (2.33) yields

$$f_x(0,y) = -\sin y, \quad f_y(0,y) = 0, \quad y \neq 0,$$

so

$$f_{xy}(0,0) = \lim_{y \rightarrow 0} \frac{f_x(0,y) - f_x(0,0)}{y} = \lim_{y \rightarrow 0} \frac{-\sin y - 0}{y} = -1,$$

$$f_{yy}(0,0) = \lim_{y \rightarrow 0} \frac{f_y(0,y) - f_y(0,0)}{y} = \lim_{y \rightarrow 0} \frac{0 - 0}{y} = 0.$$

2.16.2 Equality of Mixed Partial Derivatives

Theorem: Suppose that f , f_x , f_y , and f_{xy} exist on a neighborhood N of (x_0, y_0) and f_{xy} is continuous at (x_0, y_0) .

Then $f_{yx}(x_0, y_0)$ exists, and

$$f_{yx}(x_0, y_0) = f_{xy}(x_0, y_0).$$

Proof: Suppose that $\varepsilon > 0$. Choose $\delta > 0$ so that the open

$$S_\delta = \{(x, y) : |x - x_0| < \delta, |y - y_0| < \delta\}$$

is in N .

$$|f_{yx}(\hat{x}, \hat{y}) - f_{xy}(\hat{x}, \hat{y})| < \varepsilon \quad \text{if} \quad (\hat{x}, \hat{y}) \in S_\delta$$

A critical point at which a function attains its minimum value among all points where it is defined is called

Answer (Please select your correct option)

☐ global maximum

☐ global minimum

☐ supremum

☐ None of these

In handouts , There is no
Concept of globe max,min ,,
ther local max. and local
min. & I think D is the right
Answer

correct



2.18 Maxima and Minima

We say that $\mathbf{X}_0$ is a *local extreme point* of f if there is a $\delta > 0$ such that

$$f(\mathbf{X}) - f(\mathbf{X}_0)$$

does not change sign in $S_\delta(\mathbf{X}_0) \cap D_f$.

More specifically, $\mathbf{X}_0$ is a *local maximum point* if

$$f(\mathbf{X}) \leq f(\mathbf{X}_0)$$

or a *local minimum point* if

$$f(\mathbf{X}) \geq f(\mathbf{X}_0)$$

for all $\mathbf{X}$ in $S_\delta(\mathbf{X}_0) \cap D_f$.

A $f_1, f_2, \dots, f_k$ exist on a neighborhood of X_0 and are continuous at X_0 , then f is — at X_0

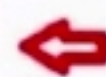
page 115



Answer (Please select your correct option)

☐ uniformly converget☐ piecewise convergent☐ differentiable☐ None of these

EDUCATION INFORMATION EXPERT



2.17.3 A sufficient Condition for Differentiability

Theorem: If $f_{x_1}, f_{x_2}, \dots, f_{x_n}$ exist on a neighborhood of $\mathbf{X}_0$ and are continuous at $\mathbf{X}_0$, then f is differentiable at $\mathbf{X}_0$.

Proof: Let $\mathbf{X}_0 = (x_{10}, x_{20}, \dots, x_{n0})$ and suppose that $\varepsilon > 0$. Our assumptions imply that there is a $\delta > 0$ such that $f_{x_1}, f_{x_2}, \dots, f_{x_n}$ are defined in the n -ball

$$S_\delta(\mathbf{X}_0) = \{\mathbf{X} : |\mathbf{X} - \mathbf{X}_0| < \delta\}$$

and

A set $A \subset \mathbb{R}$ of real numbers is _____ if there exists a real number $M \in \mathbb{R}$, such that $x \leq M$ for every $x \in A$.



Answer: (Please select your correct option)

☐ bounded below

☐ uniformly continuous

☐ bounded above

☐ None of these

May be Bounded above

or

may be the D option.. plz check



1.12 Pointwise and Uniform Bounded Functions

A sequence of functions $\{F_n\}$ on the set S is said to be pointwise **bounded** on S if the sequence of functions is **bounded** for every $x \in S$, that is, if there exists a finite valued function $\phi(x)$ defined on S such that

$$|F_n(x)| < \phi(x), \quad x \in S, n = 1, 2, 3, \dots$$

We say that $\{F_n\}$ is uniformly **bounded** on S if there exist a number M such that

$$|F_n(x)| < M, \quad x \in S, n = 1, 2, 3, \dots$$

conclude that $f(\mathbf{C}) = u$ for some $\mathbf{C}$ in S .

Theorem: A function f is **uniformly continuous** on a subset S of its domain in $\mathbb{R}^n$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(\mathbf{X}) - f(\mathbf{X}')| < \varepsilon$$

whenever

$$|\mathbf{X} - \mathbf{X}'| < \delta$$

and $\mathbf{X}, \mathbf{X}' \in S$.

Remark: We emphasize that δ must depend only on ε and S , and not on the particular points $\mathbf{X}$ and $\mathbf{X}'$.

Theorem: If f is continuous on a compact set S in $\mathbb{R}^n$, then f is **uniformly continuous** on S .

$$|x - p_1| < \delta.$$

Since $\{F_n\}$ is pointwise **bounded**, there exists $M_i < \infty$ such that

$$|F_n(p_i)| < M_i, n \in \mathbb{N}.$$

If we take

$$M = \max\{M_1, \dots, M_r\},$$

then $|F_n(x)| < M + \varepsilon$ for every $x \in S$. This proves the first part of the theorem.

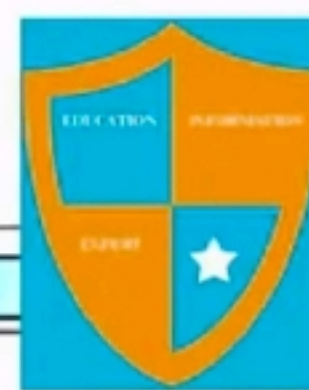
THEOREM 10.1.1. *If $\{F_n\}$ is a sequence of functions defined on a compact set*

We say that $\{F_n\}$ is uniformly bounded on S if there exist a number M such that

$$|F_n(x)| < M, \quad x \in S, n = 1, 2, 3, \dots$$

In a vector valued function $G = (g_1, g_2, \dots, g_n)$. Then $g_1, g_2, \dots, g_n$ are called the _____ functions of G .

page 114



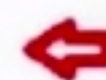
Answer (Please select your correct option)

☐ bijective

☐ surjective

☐ component

☐ None of these



2.19 Differentiable Vector Valued Function

A vector-valued function $\mathbf{G} = (g_1, g_2, \dots, g_n)$ is *differentiable* at

$$\mathbf{U}_0 = (u_{10}, u_{20}, \dots, u_{m0})$$

if its **component** functions $g_1, g_2, \dots, g_n$ are differentiable at $\mathbf{U}_0$.

Lemma: Suppose that $\mathbf{G} = (g_1, g_2, \dots, g_n)$ is differentiable at

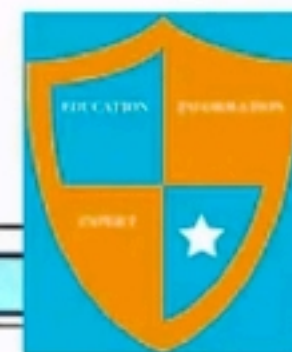
$$\mathbf{U}_0 = (u_{10}, u_{20}, \dots, u_{m0}),$$

and define

$$M = \left(\sum_{i=1}^n \sum_{j=1}^m \left(\frac{\partial g_i(\mathbf{U}_0)}{\partial u_j} \right)^2 \right)^{1/2}.$$

Then if $\epsilon > 0$ there is a $\delta > 0$ such that

The quotient of two continuous functions is a _____ function wherever the denominator is non-zero.



Answer (Please select your correct option)

☐ differentiable

☒ continuous

☐ composite

☐ None of these

EDUCATION INFORMATION EXPERT



The set $S = \{(x, y), x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 \geq 4\}$ is not a region in $\mathbb{R}^2$, since it is _____

page 90



Answer (Please select your correct option)

☐ not connected.

☒ connected.

☐ polygonally connected.

☐ None of these



Example: Intervals are the only regions in $\mathbb{R}$. The n -ball $B_r(\mathbf{X}_0)$ is a region in $\mathbb{R}^n$, as is its closure $\overline{B}_r(\mathbf{X}_0)$. The set $S = \{(x, y) : x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 \geq 4\}$ is not a region in $\mathbb{R}^2$, since it is **not connected**.

The set S_1 obtained by adding the line segment

$$L_1: \quad \mathbf{X} = t(0, 2) + (1 - t)(0, 1), \quad 0 < t < 1,$$



A set S is polygonally connected if every pair of points in S can be connected by a polygonal path lying _____



page 89

Answer (Please select your correct option)

☐ outside the set S ☐ partially in S ☐ entirely in S ☐ None of these

2.8 Polygonally Connected Set

A set S is *polygonally connected* if every pair of points in S can be connected by a polygonal path lying entirely in S .

Theorem: An open set S in $\mathbb{R}^n$ is connected if and only if it is polygonally connected.

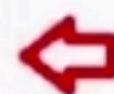
A set is closed if and only if it always contains all of its

page 86



Answer (Please select your correct option)

☐ limit points



☐ neighbourhood points

☐ interior points

☐ None of these



limit $\mathbf{X}$. Since $\mathbf{X}$ is a limit point of every S_k and every S_k is closed, $\mathbf{X}$ is in S_k (A set is **closed** if and only if it contains all its limit points). Therefore $\overline{\mathbf{X}} \in I$ so $I \neq \emptyset$. Moreover $\overline{\mathbf{X}}$ is the only point in I since if $\mathbf{Y} \in I$ then

The lower integral $\int_R f(x)dx$ of f over R is the supremum of all the

page 135



Answer (Please select your correct option)

☐ lower sums



☐ upper sums

☐ average sums

☐ None of these



$$s(\mathbf{P}) = \sum_{j=1}^k m_j V(R_j).$$

The *lower integral of f over R* , denoted by

$$\underline{\int_R} f(\mathbf{X}) \, d\mathbf{X},$$

is the supremum of all lower sums.

Theorem: Let f be bounded on a rectangle R and let

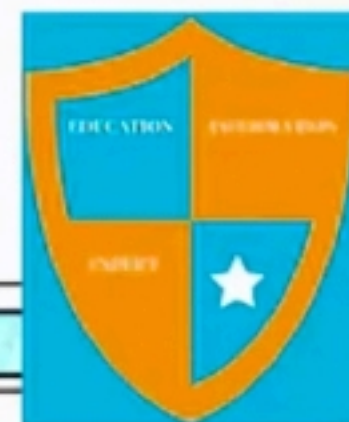
Then

1. The upper sum $S(\mathbf{P})$ of f over $\mathbf{P}$ is the supremum of all upper sums of f over $\mathbf{P}$.

2. If f is Riemann integrable on R , then $\underline{\int_R} f(\mathbf{X}) \, d\mathbf{X} = \int_R f(\mathbf{X}) \, d\mathbf{X} = \overline{\int_R} f(\mathbf{X}) \, d\mathbf{X}$.

If f and g are integrable on S and $f(X) \leq g(X)$ for X in S , then ———

page163



Answer (Please select your correct option)

☐ $\int_S f(X) dX \geq \int_S g(X) dX$

☒ $\int_S f(X) dX \leq \int_S g(X) dX$

☐ $\int_S f(X) dX > \int_S g(X) dX$

☐ None of these



Theorem: If f and g are integrable on S and $f(\mathbf{X}) \leq g(\mathbf{X})$ for $\mathbf{X}$ in S , then

$$\int_S f(\mathbf{X}) d\mathbf{X} \leq \int_S g(\mathbf{X}) d\mathbf{X}.$$

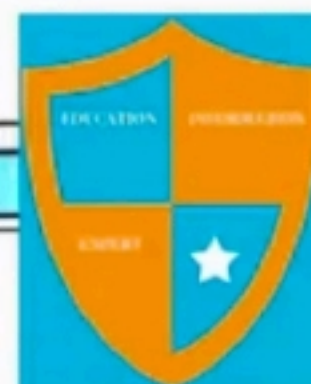
Theorem: If f is integrable on S , then so is $|f|$, and

$$\left| \int_S f(\mathbf{X}) d\mathbf{X} \right| \leq \int_S |f(\mathbf{X})| d\mathbf{X}.$$

Theorem: If f and g are integrable on S , then so is the product fg .

The upper integral $\overline{\int}_a^b f(x)dx$ of f over R is the infimum of all the _____

page 135



Answer (Please select your correct option)

☐ lower sums☒ upper sums☐ average sums☐ None of theseEDUCATION INFORMATION
EXPERT

3.9 Upper and Lower Integrals

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \dots, R_k\}$ is a partition of R .

Let

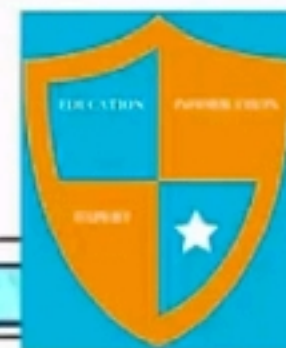
$$M_j = \sup_{\mathbf{X} \in R_j} f(\mathbf{X}), \quad m_j = \inf_{\mathbf{X} \in R_j} f(\mathbf{X}).$$

The *upper sum* of f over $\mathbf{P}$ is

$$S(\mathbf{P}) = \sum_{j=1}^k M_j V(R_j).$$

A function f is said to be absolutely integrable on $[a, b]$ if f is locally integrable on $[a, b]$ and ———

page 145



Answer (Please select your correct option)

☐ $\int_a^b |f(x)| dx < \infty$



☐ $\int_a^b |f(x)| dx = \infty$

☐ $\int_a^b |f(x)| dx \geq \infty$

☐ None of these

3.2 Absolute integrability

We say that f is *absolutely integrable* on $[a, b)$ if f is locally integrable on $[a, b)$ and $\int_a^b |f(x)| dx < \infty$. In this case we also say that $\int_a^b f(x) dx$ *converges absolutely* or is *absolutely convergent*.

Remark: If f is nonnegative and integrable on $[a, b)$, then f is *absolutely integrable* on $[a, b)$, since $|f| = f$.

Example: The empty set and **singleton** sets are connected, because they cannot be represented as the union of two disjoint nonempty sets.

2.6 Heine-Borel Theorem

We are going to state and prove the Heine-Borel theorem for $\mathbb{R}^n$.

This theorem concerns *compact* sets. As in $\mathbb{R}$, a compact set in $\mathbb{R}^n$ is a closed and **bounded** set.

of *supremum* and *infimum* of a function on a set S .

Theorem: If f is continuous on a compact set S in $\mathbb{R}^n$, then f is **bounded** on S .